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Momentum project and the assessment of human movement: accessibility, low cost, and technological challenges

Projeto Momentum e a avaliação do movimento humano: acessibilidade, baixo custo e desafios tecnológicos

El proyecto Momentum y la evaluación del movimiento humano: la accesibilidad, el bajo costo y los desafíos tecnológicos

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Lately, the use of smartphones to assess human movement has been shown as an innovative and promising approach, especially in developing countries. Equipped with accelerometers, gyroscopes, magnetometers, cameras, microphone, and touch screen, smartphones represent an affordable and low-cost solution when compared to traditional motor assessment methods, such as force platforms and motion capture systems, which require high investments and specialized infrastructure. The growing popularity of smartphones in different populational strata makes them a valuable tool for functional monitoring in contexts with limited resources.

Several studies have investigated the use of smartphones for motor monitoring, confirming their applicability in rehabilitation, balance assessment, and early detection of motor impairments. Our group at the Federal University of Pará has developed mobile applications (*Momentum Science* and *Momentum Touch*), developed motor assessment protocols using these apps as a tool for recording signals related to human movement, and shown that the extracted data are valid when compared to gold-standard methods. These tools have significant replicability, can differentiate healthy subjects from those with different clinical conditions, and enable assessing factors that may interfere with the performance of the tests. *Momentum Science* is a mobile application for recording signals from the accelerometer and gyroscope built into the smartphone. *Momentum Touch* records time and space data of touches on the smartphone screen.

Inertial signals obtained via the *Momentum Science* app during static balance tasks and the evaluation of anticipatory postural adjustments were similar to those obtained via the force platform and motion capture system in healthy young adults[1]. Inertial signs have been studied in patients with diabetes mellitus, long COVID, leprosy, and osteoarthritis, presenting disease-related functional losses[2,3]. We also tested the replicability of inertial records during the timed up and go test, achieving high replicability when the test was carried out using the same smartphone or different ones[4]. Using the *Momentum Touch* app, which recorded touch on the smartphone screen, we showed that sex and handedness could bring significant differences in the performance of the finger tapping test, drawing attention to the relevance of considering these two factors during clinical evaluation using this type of test. We also evaluated the characteristics of hand trembling, using lightweight wearable sensors and *Momentum Science,* and observed that the data are not comparable and the weight of the device (a few or hundreds of grams) modifies the spectral characteristics of the inertial signals related to hand trembling.

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Despite the obvious advantages, technological challenges still need an overcome. The main obstacle is still the development of signal processing algorithms that can translate the information collected by *Momentum Sensors* into intuitively actionable clinical data. Healthcare providers need tools that not only collect data, but also make it easier to interpret such data for clinical decision-making. The creation of simple and user-friendly interfaces, which can be used by professionals with different levels of training, is an important step towards the widespread adoption of such technologies. Furthermore, there are variations between devices and conditions of use that can influence measurements accuracy, making the development of standardized protocols and the continuous validation of these tools crucial. For example, different smartphone models may have variations in the sensitivity of their sensors, which may impact data consistency, especially in clinical settings that deeply rely on accuracy .

The apps *Momentum Science* and *Momentum Touch* are still under development and in experimental phase, and we have been evaluating the implementation of solutions that allow their use outside the research laboratory and in real-life settings. Currently, the applications are being used in final papers, theses, and dissertations at the Federal University of Pará, the State University of Pará, the Federal University of the State of Pará, the Federal University of Amapá, and the Federal University of Roraima. These mobile apps represent innovative and affordable tools for the assessment of human movement, with multiple clinical uses already validated and an increasing potential to improve access to healthcare, especially in regions with limited resources. However, so this technology can reach its maximum potential, it is necessary to keep financing the improvement of data processing algorithms, as well as the creation of intuitive interfaces that facilitate clinical use. Ongoing validation of these tools will also be key to ensure that the obtained data is accurate and usable, enabling healthcare providers to make informed and personalized decisions in their patients' care.

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